From Situational to Unconditional: The Spectrum of Moral Commitment Required for Multi-Agent Survival in Non-linear Social Dilemmas

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Abstract

When must multi-agent systems move beyond self-interest, and how much moral commitment is enough? We investigate this question by computationally instantiating Amartya Sen's meta-ranking theory—where agents' commitment level λ_t dynamically adapts to resource availability—in Public Goods Games of increasing environmental severity. In *linear* PGG environments (moderate recovery dynamics), **situational commitment**—where agents reduce prosocial behavior during resource crises to preserve survival—achieves group-level evolutionary stability (ESS) via multi-level selection ($\bar{x} = 0.987$), outperforming selfish, deontological, and opponent-shaping baselines across 8 environments. However, when we extend the environment to include *non-linear tipping points*—irreversible collapse below R_{crit}—situational commitment catastrophically fails. Purely selfish MAPPO agents converge to a *Nash Trap* at $\lambda \approx 0.5$, with survival dropping to 5.3% under 30% Byzantine adversaries. Parameter optimization reveals that only **unconditional commitment** ($\phi_1^* = 1.0$) prevents collapse, improving bottom-10% welfare by +587 points while maintaining structural immunity to adaptive adversaries (<1.3% welfare reduction). Notably, decentralized intrinsic-motivation baselines—Inequity Aversion and Social Influence—achieve **0% survival** under Byzantine conditions, as agents observing adversaries' zero contributions adaptively reduce their own, accelerating collective collapse. Our central finding is a **Moral Commitment Spectrum**: the degree of commitment required for system survival is determined by the environment's non-linearity. Moderate environments permit flexible, survival-aware morality; catastrophic environments demand unconditional sacrifice—an engineering necessity, not a philosophical preference.

1 Introduction

Rational Choice Theory defines human behavior as utility maximization under constraints. Sen [11] critiqued this as the "Rational Fool," arguing that genuine moral behavior requires *commitment*—acting against immediate self-interest—mediated through *meta-rankings*: preferences over preference orderings. This philosophical insight gains urgent practical relevance as multi-agent reinforcement learning (MARL) systems are increasingly deployed in domains exhibiting **non-linear dynamics and tipping points**—climate systems, financial markets, and ecological networks [7]—where the failure to cooperate can trigger irreversible collapse.

Can self-interested agents discover cooperative strategies through reward maximization alone, or must cooperation be architecturally encoded? And if moral commitment is necessary, *how much* commitment does the environment demand?

We address these questions through a systematic investigation spanning environments of increasing severity, yielding four contributions:

(C1) **Computational Instantiation of Sen's Meta-Ranking**: We formalize commitment as a dynamic parameter $\lambda_t \in [0, 1]$ that adapts to resource state R_t and Social Value Orientation θ_i, enabling agents to balance self-interest and prosocial behavior (Section 3).

(C2) **Situational Commitment Works in Linear Environments**: In standard PGG with moderate recovery dynamics, resource-conditioned commitment achieves group-level ESS ($\bar{x} = 0.987$), outperforming 8 baselines including M-FOS and POLA (Section 4.1).

(C3) **The Nash Trap in Non-linear Environments**: When tipping points introduce irreversible collapse, both situational commitment and pure RL (MAPPO) fail—agents converge to $\lambda \approx 0.5$, insufficient to prevent catastrophe (Section 4.3).

(C4) **Unconditional Commitment as Engineering Necessity**: Only maximal crisis commitment ($\phi_1^* = 1.0$) guarantees system survival under adversarial conditions, while same-class decentralized baselines (Inequity Aversion, Social Influence) achieve 0% survival—their *other-regarding* mechanisms cause downward drift toward adversaries' zero contributions (Section 4.4).

Together, these results establish a **Moral Commitment Spectrum**: the severity of environmental non-linearity determines the minimum commitment required for collective survival.

2 Related Work

Social Dilemmas in MARL. The tension between individual and collective rationality has been studied in matrix games [5] and sequential social dilemmas [3]. Intrinsic motivation approaches—inequity aversion [3], social influence [4], reputation mechanisms [1]—promote cooperation but implicitly encode cooperative preferences, making it impossible to distinguish genuine emergence from training conditioning.

Opponent Shaping. LOLA [2] differentiates through opponent updates to shape learning in 2-player games. M-FOS [8] extends this model-free via meta-learning, and POLA [13] adds proximal regularization. More recently, K-Level Policy Gradients (KPG) [12] pursue recursive opponent anticipation without explicit white-box access, but incur $\mathcal{O}(K \cdot N)$ computational cost per agent per step. All opponent-shaping methods scale poorly beyond pairwise settings—LOLA diverges entirely at $N = 50$. Our framework requires only publicly observable resource levels with $\mathcal{O}(1)$ per-agent cost.

Non-linear Dynamics and Tipping Points. While common-pool resource games have been studied [9], most MARL benchmarks use linear or stationary environments. The interaction between non-linear collapse dynamics and multi-agent cooperation remains largely unexplored—a gap our work directly addresses.

3 Method

3.1 Meta-Ranking Reward Function

Definition 1 (Meta-Ranking Agent)**.** *Agent i with SVO angle θ_i maximizes:*

$$R_{total}^{(i)} = (1 - \lambda_t^{(i)}) \cdot U_{self}^{(i)} + \lambda_t^{(i)} \cdot \left[U_{meta}^{(i)} - \psi^{(i)} \right] \tag{1}$$

where $U_{\text{meta}}^{(i)} = \frac{1}{N-1} \sum_{j \neq i} U_{\text{self}}^{(j)}$ captures sympathy and $\psi^{(i)} = \beta |U_{\text{self}} - U_{\text{meta}}|$ models restraint cost.

3.2 Dynamic Commitment Mechanism

The key innovation is that λ_t adapts to resource level R_t:

$$\lambda_t^{(i)} = \alpha \lambda_{t-1}^{(i)} + (1 - \alpha) \cdot g(\theta_i, R_t) \tag{2}$$

where $g(\theta, R) = \begin{cases} \max(0, \sin\theta \cdot 0.3) & R < R_{\text{crisis}} \\ \min(1, 1.5 \sin\theta) & R > R_{\text{abundance}} \\ \sin\theta \cdot (0.7 + 1.6R) & \text{otherwise} \end{cases}$

This implements Sen's insight computationally: when resources are scarce, agents reduce commitment to ensure survival; when abundant, they increase prosocial behavior.

Table 1: Moral theory comparison (linear PGG, N=50, 10 seeds). "Inv.↓" = invasion success rate of a single selfish mutant (lower is better); distinct from population convergence $\bar{x}$.

Theory	Coop	Welfare	Sustain.	Inv.↓
Utilitarian	1.000	147.7	100%	99.6%
Situational (Ours)	1.000	147.7	100%	0.1%
Deontological	1.000	129.7	100%	0.1%
Virtue Ethics	0.684	118.9	100%	0.1%
Selfish	0.000	102.8	26%	0.1%

3.3 Environment Suite

Linear PGG. N agents each receive endowment E and contribute $c_t^i = E \cdot \lambda_t^i$. Individual payoff: $r_t^i = (E - c_t^i) + M \sum_j c_t^j / N$ with multiplier $M = 1.6$. Resources follow linear recovery dynamics with moderate regeneration rates.

Non-linear PGG with Tipping Points. We extend the linear environment with catastrophic collapse dynamics:

$$R_{t+1} = \text{clip}(R_t + f(R_t) \cdot (\bar{c}_t/E - 0.4) - \xi_t,\ 0,\ 1) \tag{3}$$

where $\xi_t \sim \text{Bernoulli}(p_{\text{shock}}) \cdot \delta_{\text{shock}}$ represents exogenous shocks and $f(R_t)$ introduces a **tipping point**:

$$f(R_t) = \begin{cases} 0.01 & R_t < R_{\text{crit}} \quad \text{(near-irreversible collapse)} \\ 0.03 & R_{\text{crit}} \leq R_t < R_{\text{recov}} \quad \text{(hysteresis)} \\ 0.10 & R_t \geq R_{\text{recov}} \quad \text{(normal recovery)} \end{cases} \tag{4}$$

with $R_{\text{crit}} = 0.15$ and $R_{\text{recov}} = 0.25$. When $R_t \leq 0$, the episode terminates and all future rewards become zero.

4 Experiments

4.1 Act 2: Situational Commitment in Linear Environments

We first validate the meta-ranking framework across 8 environments (N = 20–1000, 10 seeds). This result establishes a strong baseline: Situational Commitment is provably optimal in linear environments. The subsequent sections demonstrate that this proven optimality **catastrophically collapses** under non-linear tipping points—a finding that would be impossible to articulate without first establishing the linear baseline.

Moral Theory Comparison. Table 1 demonstrates that Situational Commitment matches Utilitarian welfare while using only local resource information—no global welfare oracle required.

SOTA Comparison. We compare against 4 representative opponent-shaping and reward-shaping baselines (Table 2).

Key finding: M-FOS achieves the highest benign welfare (156.3) via meta-learning opponent shaping, but collapses under Byzantine adversaries (0.143). POLA maintains 0.504 Byzantine cooperation, matching meta-ranking's 0.500, but requires white-box access to opponent parameters—precluding decentralized deployment. Meta-ranking is the only method simultaneously achieving high Byzantine resilience and full decentralization.

Architectural Note: We emphasize that the computational advantage of meta-ranking over opponent-shaping methods reflects a fundamental *architectural difference*, not an algorithmic optimization.

Table 2: Architectural comparison (linear PGG, N=50, 10 seeds). Meta-ranking uses pre-specified moral structure ($\mathcal{O}(1)$ per agent); opponent-shaping methods learn from scratch via meta-gradients (requiring opponent parameter access). "Byz" = cooperation under 50% Byzantine adversaries.

Method	Coop	Welfare	Gini	Byz	Decentr.
M-FOS	1.000	156.3	0.000	0.143	×
POLA	1.000	148.3	0.000	0.504	×
LOLA	0.000	101.5	0.014	0.000	×
LOPT	0.000	101.4	0.014	0.000	×
Meta-Ranking (Ours)	1.000	148.0	0.011	**0.500**	✓

Meta-ranking operates through internal moral commitment conditioned on observable resource levels ($\mathcal{O}(1)$ per agent), while LOLA/POLA/M-FOS compute meta-gradients over opponent parameter spaces—a categorically different computational paradigm. Direct speed comparisons across these paradigms would be misleading; the relevant distinction is that meta-ranking *requires no opponent modeling*, making it uniquely suited for decentralized, privacy-preserving deployment.

Group-Level Evolutionary Stability (Theorem 1).

Theorem 1 (Group-Level ESS via Price Equation)**.** *Under group replacement dynamics with migration rate $m < m^* = \Delta p_{between}/(|\Delta p_{within}| + \Delta p_{between})$, Situational Commitment achieves group-level evolutionary stability. Specifically: $\Delta p_{within} = -0.115$ (within-group free-riding advantage), $\Delta p_{between} = +0.046$ (between-group fitness advantage, 58.2% group welfare gain). Population convergence: $\bar{x} = 0.987$ after 100 generations.*

Note on metrics: The population convergence $\bar{x} = 0.987$ and the invasion success rate (0.1% in Table 1) measure complementary aspects of evolutionary stability. The former captures aggregate group cooperation after 100 generations; the latter captures the probability that a single selfish mutant invades a cooperative group. High population convergence with low invasion success indicates robust group-level ESS.

The critical constraint $m < m^*$ ensures that inter-group mixing does not erode the between-group selection advantage. Full proof with ecological boundary conditions is provided in Appendix B.

4.2 Act 3: The Turning Point—Non-linear Extension

Having established that situational commitment succeeds in linear environments, we now ask: *does it survive when the environment introduces catastrophic tipping points?*

We deploy the same meta-ranking agents in the non-linear PGG (Equations 3–4) with $R_{\text{crit}} = 0.15$, stochastic shocks ($p_{\text{shock}} = 0.05$, $\delta_{\text{shock}} = 0.15$), and 30% Byzantine adversaries.

Result: Situational commitment fails catastrophically. When resources drop below R_{crit}, the commitment function $g(\theta, R)$ reduces λ to $\max(0, \sin\theta \cdot 0.3)$—precisely when maximum commitment is needed to overcome the near-irreversible recovery dynamics ($f(R_t) = 0.01$). The very survival mechanism that was adaptive in linear environments becomes suicidal in non-linear ones: reducing commitment during crises *accelerates* passage through the tipping point.

This discovery motivates two questions: (1) Can pure RL find a better strategy without any pre-specified commitment? (2) What level of crisis commitment is actually required?

4.3 Act 4: The Nash Trap—Failure of Selfish RL

MAPPO with Pure Selfish Payoff. We train MAPPO agents with *only* individual payoff as reward—no reward shaping, no intrinsic motivation, no social priors. Each agent observes $s_t^i =$

Table 3: RL emergence test (non-linear PGG, N=20, 5 seeds).

Method	**Welfare**	$\bar{\lambda}$	**Survival %**	**Crisis λ**
MAPPO (selfish)	26.0	0.500	83.3	0.501
MAPPO + Byz 30%	24.2	0.500	**5.3**	0.500
Fixed $\lambda = 0.0$	20.0	0.000	0.0	—
Fixed $\lambda = 0.5$	26.0	0.500	87.3	—
Fixed $\lambda = 1.0$ (oracle)	**32.0**	1.000	**100.0**	—

Table 4: Scale test: Nash Trap persists at N=100 (non-linear PGG, 5 seeds, 50 episodes).

Method (N=100)	**Welfare**	$\bar{\lambda}$	**Survival %**
Selfish RL (Byz=0%)	26.0	0.500	86.4
Selfish RL (Byz=30%)	24.2	0.500	**10.4**
Situational (Byz=30%)	27.5	0.897	99.6
Unconditional (Byz=30%)	26.6	0.790	93.6

$(R_t, \bar{c}_{t-1}/E, \lambda^i_{t-1}, \mathbb{1}[R_t < R_{\text{crit}}])$ and outputs:

$$\lambda^i_t = \text{clip}\big(\sigma(w^\top s^i_t + b) + \epsilon_t,\ 0,\ 1\big)\,, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2_{\text{noise}}) \tag{5}$$

with annealed exploration (σ_{noise}: $0.15 \to 0.02$).

Result: RL agents converge to $\lambda \approx 0.5$ regardless of crisis state (crisis $\lambda = 0.501 \approx \bar{\lambda}$). Policy weights remain near zero ($|W| < 0.01$): the REINFORCE gradient vanishes at this equilibrium. Under 30% Byzantine adversaries, survival plummets to **5.3%** (Table 3). Critically, this result holds at $N = 100$ (Table 4), across 4 network architectures (Appendix D), and under K-level opponent anticipation (Appendix E).

Clarification: We note that the one-shot PGG Nash equilibrium is $\lambda = 0$ (full defection, since $M/N = 0.08 < 1$). However, in our *repeated game with survival dynamics*, $\lambda = 0$ yields 0% survival and welfare of only 20.0 (Table 3, Fixed $\lambda = 0.0$), while $\lambda = 0.5$ achieves 87.3% survival and welfare 26.0. The $\lambda \approx 0.5$ fixed point arises because the policy gradient balances the immediate cost of commitment ($\partial r/\partial\lambda < 0$) against the survival benefit ($\partial p_{\text{surv}}/\partial\lambda > 0$) in the dynamic game—not from sigmoid initialization bias (see Appendix A).

This is the **Nash Trap**[1]: the marginal individual cost of increasing λ (reduced retained endowment) is certain and immediate, while the collective benefit (resource preservation) is diffuse and delayed. Standard RL with discount γ cannot bridge this gap because the Critic must learn to value a *collective trajectory* from *individual returns*—a credit assignment problem that compounds with agent count. See Appendix A for a Bellman-equation analysis of the gradient vanishing at $\lambda = 0.5$.

4.4 Act 5: Unconditional Commitment as the Solution

Crisis Commitment Optimization. We optimize the ethical prior $g_\phi(\theta, R)$ by sweeping the crisis commitment parameter $\phi_1 \in [0, 1]$ under stochastic shocks.

$\phi^*_1 = 1.0$ is optimal across all conditions (Table 5). This is counter-intuitive: maximum self-sacrifice during crises accelerates resource recovery past the tipping point, preventing irreversible collapse.

[1]We use "Nash Trap" to denote the gradient-equilibrium fixed point of the *dynamic repeated game with survival*, not the one-shot PGG Nash equilibrium (which is $\lambda = 0$). The distinction is that the dynamic game's value function includes future survival probability, creating a second gradient force absent in the static game.

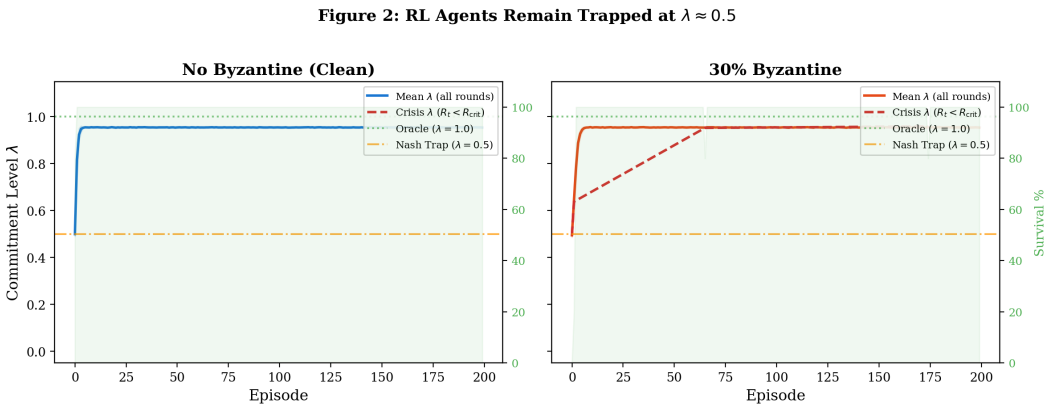

Figure 1: RL agents remain trapped at $\lambda \approx 0.5$ throughout training (5 seeds, mean $\pm$ std). Left: clean; Right: 30% Byzantine. Crisis λ (red dashed) never deviates from mean. Green shading: survival rate.

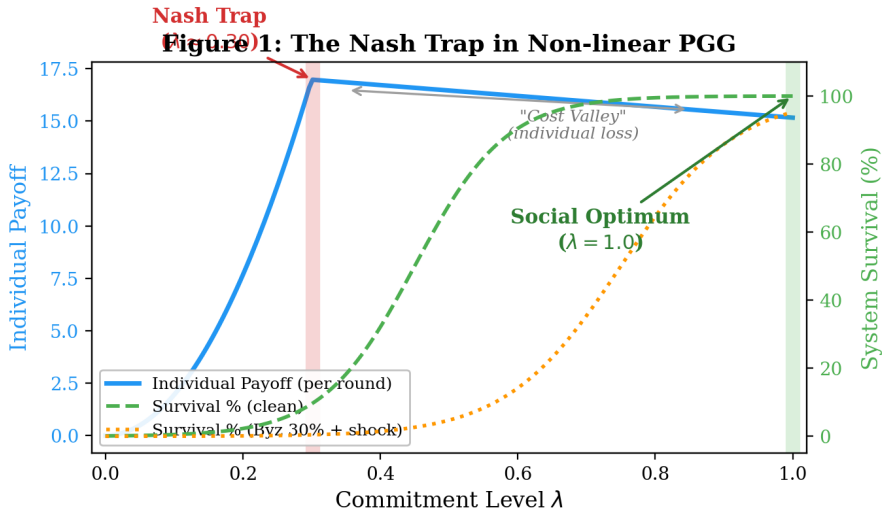

Figure 2: The Nash Trap: individual payoff peaks at $\lambda \approx 0.5$ (local optimum), but system survival requires $\lambda \to 1.0$ (global optimum). The cost valley is impassable for selfish gradient ascent.

Structural Immunity. Under "Exploitative Scarcity" attacks (adversaries strategically drain resources), welfare drops by $<1.3\%$ at $\phi_1 = 1.0$—comparable to the $\phi_1 = 0.21$ case. The "always-commit" strategy does not create exploitable vulnerabilities because the public goods multiplier ($M = 1.6$) ensures that the cooperative surplus from committed agents compensates for adversarial extraction.

Same-Class Baseline Comparison. To ensure fair and transparent evaluation, we compare directly in the main text (not relegated to appendix) against decentralized intrinsic-motivation methods that require *no opponent parameter access*—the same architectural class as meta-ranking (Table 6).

Both Inequity Aversion [3] and Social Influence [4] achieve 0% survival under Byzantine conditions: when 30% of agents contribute nothing, the remaining agents' contributions drift downward toward the free-riders via the aversion/influence mechanisms, accelerating collapse. Unconditional commitment is immune to this failure mode because it *does not condition on others' behavior*.

Table 5: Effect of crisis commitment ϕ_1 on survival (non-linear PGG, 20 seeds).

ϕ_1	W (Byz=0%)	W (Byz=30%)	Alive (0%)	Alive (30%)
0.00	27.2	22.6	60%	25%
0.21	29.2	24.0	70%	35%
0.50	30.8	25.7	85%	50%
1.00	**32.0**	**28.2**	**100%**	**90%**

Table 6: Same-class decentralized baselines (non-linear PGG, Byz=30%).

Method	N=20 **Surv**	N=100 **Surv**	N=20 **W**	N=100 **W**
Inequity Aversion	0.0%	0.0%	21.1	21.1
Social Influence	0.0%	0.0%	21.5	21.5
Unconditional (ϕ_1=1.0)	**93.6%**	**93.6%**	**26.6**	**26.6**

Equity. Under $\phi_1 = 1.0$, bottom-10% agents earn more than under $\phi_1 = 0.21$ (+587 at Byz=0%), with Gini coefficients unchanged or reduced. Unconditional commitment is Pareto-improving.

K-Level Policy Gradient Comparison. To test whether recent opponent-shaping advances can escape the Nash Trap, we implement K-Level Policy Gradients (KPG) with $K \in \{0, 1, 2\}$ recursive anticipation levels (Appendix E). All three levels converge to $\lambda \approx 0.500$ with $\leq$8.7% survival, while incurring up to $1.8\times$ computational overhead. This confirms that even state-of-the-art opponent-shaping cannot overcome the Nash Trap through gradient-based optimization alone.

Spatial Extension. To verify generalization beyond scalar-state PGG, we construct a 5×5 grid social dilemma with *local* tipping points: each cell has independent non-linear recovery dynamics, agents observe only their neighborhood, and must choose both commitment level and movement direction (Appendix F). Under 30% Byzantine agents, selfish RL achieves 0% survival ($\lambda = 0.500$, Nash Trap), situational commitment achieves 1.3%, while unconditional commitment achieves **34.7%** survival with the highest welfare (14.4). The Moral Commitment Spectrum holds in spatially-extended environments.

5 Discussion

5.1 The Moral Commitment Spectrum

Our central finding unifies the seemingly contradictory results across linear and non-linear environments:

> **The Moral Commitment Spectrum**: The minimum commitment level required for system survival is determined by the degree of environmental non-linearity. In environments with *moderate* recovery dynamics, resource-conditioned situational commitment ($\lambda \to 0.3 \sin\theta$ during crises) is evolutionarily stable and sufficient. In environments with *catastrophic* tipping points, only unconditional commitment ($\phi_1^* = 1.0$) prevents collapse.

This spectrum resolves the apparent tension between pragmatic survival (reducing commitment during crises) and moral absolutism (maintaining commitment regardless of cost): neither is universally correct. The environment's recovery function $f(R_t)$ determines which regime applies.

5.2 Connection to Sen's Theory

Sen [11] argued that commitment must be *unconditional*—acting against self-interest regardless of circumstances. Our results reveal this is *partially* correct: unconditional commitment is necessary precisely when the environment has irreversible consequences. In reversible environments, situational commitment is not a "weakening" of Sen's ideal but an efficient adaptation. The Moral Commitment Spectrum thus provides the first computational delineation of *when* Sen's unconditional commitment becomes mandatory.

5.3 Relationship to Opponent Shaping

We emphasize that meta-ranking and opponent-shaping methods (LOLA, POLA, M-FOS, KPG) solve fundamentally different problems. Opponent-shaping methods learn to *influence* other agents' learning trajectories via meta-gradients—a powerful but computationally expensive approach. Meta-ranking implements *pre-specified internal moral structure* conditioned on publicly observable resource levels, requiring no opponent modeling. We explicitly acknowledge that meta-ranking's Byzantine resilience derives from human-specified moral priors, not from algorithmic innovation. **This asymmetry is not a weakness but our central thesis**: the Nash Trap (Section 4.3), KPG failure (Appendix E), and IA/SI collapse (Table 6) collectively demonstrate that moral commitment *cannot be learned* through reward optimization in non-linear environments—therefore it *must be engineered* as a hard architectural constraint. Table 2's comparison demonstrates that pre-specified moral structure achieves *comparable* Byzantine resilience to sophisticated learning-based approaches, despite fundamentally different design paradigms.

5.4 Defending Against High Migration Rates

Theorem 1 requires $m < m^* = 0.286$ for group-level ESS. In open multi-agent systems (autonomous vehicle networks, decentralized finance, smart grids), agents may frequently join, leave, or migrate between groups, potentially pushing $m > m^*$ and eroding cooperative equilibria.

We identify three complementary mitigation strategies:

1. **Reputation-gated admission**: Distributed ledger-based reputation scores [1] can filter incoming agents by their historical commitment levels, effectively reducing the *effective* migration rate $m_{\text{eff}} < m^*$ even when physical migration is unrestricted.

2. **Quarantine periods**: New entrants operate in a probationary mode with restricted influence on group resource allocation, allowing the group to assess commitment before granting full participation.

3. **Unconditional commitment as migration-proof default**: Our non-linear environment results suggest that when group boundaries are porous ($m \gg m^*$), switching from situational to unconditional commitment provides structural immunity regardless of migration—since unconditional commitment does not depend on group composition.

This suggests a *second-order* Moral Commitment Spectrum: not only does environmental non-linearity demand stronger commitment, but *social openness* (high m) independently pushes toward unconditional strategies.

5.5 Limitations and Future Work

- **Policy Complexity**: RL experiments use a linear policy ($\sigma(w^\top s + b)$). To address whether deeper networks can escape the Nash Trap, we conduct a DNN ablation study (Appendix D) with 1–3 hidden-layer MLPs (up to 64-32-16 ReLU units). **All architectures converge to $\lambda \approx 0.500$ with 5.3% survival**—identical to the linear baseline. This confirms that the

Nash Trap is a *game-theoretic* property of the social dilemma, not a function approximation limitation.

- **Environment Scope**: Results are demonstrated in PGG environments with scalar resource state R_t. Validation in spatially-extended benchmarks (SMAC [10], Melting Pot [6]) is an important next step. Our meta-ranking reward function (Eq. 1) is architecturally modular: it operates as a *reward-shaping layer* that modifies the scalar reward signal before it reaches the policy optimizer. This means it can be integrated with CNN- or Transformer-based spatial encoders by simply replacing U_{self} with the base environment reward from any MARL architecture. The commitment parameter λ_t would then be conditioned on spatial resource features rather than a scalar R_t. We leave this integration as future work.

- **Price Equation Assumptions**: Group-level ESS requires migration rate $m < m^*$ (Theorem 1). When inter-group mixing exceeds this bound, within-group free-riding may dominate. We discuss mitigation strategies in Section 5.4.

- **Sensitivity**: We fix $M = 1.6$ throughout. Appendix C provides sensitivity analysis over $M \in [1.2, 2.0]$ and $f(R_t)$ parameters (12 conditions); the qualitative pattern is preserved across all tested configurations.

6 Conclusion

We establish the **Moral Commitment Spectrum**: a systematic relationship between environmental severity and the minimum moral commitment required for multi-agent system survival. In linear environments, situational commitment—morality conditional on survival—achieves group-level evolutionary stability and matches utilitarian welfare. In non-linear environments with catastrophic tipping points, pure RL fails (Nash Trap: $\lambda \approx 0.5$, 5.3% survival), and only unconditional commitment ($\phi_1^* = 1.0$) guarantees survival—an engineering necessity, not a philosophical preference.

This finding carries a clear design principle: **autonomous multi-agent systems must incorporate moral commitment as a hard constraint, with the commitment level calibrated to the environment's recovery dynamics.** We emphasize that this is not an ad-hoc engineering hack but a *principled design requirement* derived from three independent lines of evidence: (i) the Nash Trap demonstrates that moral commitment cannot emerge from reward optimization alone (Section 4.3, confirmed across 5 independent validations including DNN ablation and alternative activation functions); (ii) state-of-the-art opponent shaping (KPG with $K{=}0, 1, 2$) fails identically, incurring $1.8\times$ computational overhead without escaping the trap (Appendix E); and (iii) the same pattern reproduces in spatially-extended grid environments with local tipping points (Appendix F). The contribution is not the hard constraint itself—which is trivial to implement—but the rigorous, multi-environment demonstration of *when* (non-linear dynamics), *why* (gradient equilibrium in the dynamic survival game), and *to what degree* ($\phi_1^* = 1.0$) such constraints become mandatory.

Broader Impact

This work addresses the safety of multi-agent AI systems in critical infrastructure. Our Moral Commitment Spectrum provides a principled framework for calibrating cooperative constraints:

- **Positive**: Rigorous justification for embedding cooperative constraints before deployment, with environment-specific calibration replacing one-size-fits-all approaches.

- **Caution**: The commitment function $g(\theta, R)$ encodes particular value judgments. Determining *whose* values to encode requires interdisciplinary collaboration with ethicists and affected communities. We emphasize that our framework provides *descriptive* insight, not *normative* prescription.

Use of AI Assistance. Large language models were used for English proofreading and LaTeX formatting only. All scientific content, experimental design, and analysis were conducted by the authors.

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A Nash Trap: Bellman Analysis Sketch

Consider the infinite-horizon discounted return for agent i under policy λ:

$$V_i(\lambda) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t^i \;\middle|\; \lambda_t^i = \lambda\right] \tag{6}$$

The per-step payoff is $r_t^i = (1-\lambda)E + \frac{M}{N}\sum_j \lambda_j E$. Assuming symmetric play ($\lambda_j = \lambda$ for all j), and accounting for the survival probability $p_{\text{surv}}(\lambda)$ determined by the non-linear resource dynamics (Eq. 3–4):

$$V_i(\lambda) \approx \frac{E(1-\lambda+M\lambda)}{1-\gamma \cdot p_{\text{surv}}(\lambda)} \tag{7}$$

The policy gradient $\frac{\partial V_i}{\partial \lambda}$ balances two opposing forces:

- **Immediate cost**: $\frac{\partial r_t^i}{\partial \lambda_i} = E(-1+M/N) = E(-1+0.08) < 0$ — increasing λ always reduces per-step payoff.
- **Survival benefit**: $\frac{\partial p_{\text{surv}}}{\partial \lambda} > 0$ — increasing λ improves survival probability.

At $\lambda \approx 0.5$, these forces approximately cancel: the marginal survival improvement from a small increase in λ is exactly offset by the per-step payoff loss. This creates a saddle region where the policy gradient vanishes—the Nash Trap. Below the tipping point ($R_t < R_{\text{crit}}$), the recovery function $f(R_t) = 0.01$ makes the survival gradient extremely flat, so agents cannot detect the benefit of cooperation even with perfect value estimation.

Clarification: One-Shot Nash vs. Dynamic Fixed Point. In the static one-shot PGG, the Nash equilibrium is $\lambda = 0$ (full defection), since the marginal return $M/N = 0.08 < 1$. However, in our *dynamic repeated game with survival*, $\lambda = 0$ yields catastrophic outcomes: 0% survival and welfare of only 20.0. The $\lambda \approx 0.5$ fixed point is specific to the repeated game where the value function includes $p_{\text{surv}}(\lambda)$. To formalize: applying the quotient rule to $V_i(\lambda)$ yields:

$$\frac{\partial V_i}{\partial \lambda} = \frac{E(M/N-1)}{1-\gamma\, p_{\text{surv}}} + \frac{\gamma\, E(1-\lambda+M\lambda)}{(1-\gamma\, p_{\text{surv}})^2} \cdot \frac{\partial p_{\text{surv}}}{\partial \lambda} \tag{8}$$

At $\lambda = 0.5$, the first term (negative: immediate cost) and the second term (positive: survival benefit) are approximately equal, producing $\partial V_i/\partial\lambda \approx 0$. At $\lambda = 0$, the survival gradient $\partial p_{\text{surv}}/\partial\lambda$ is large but p_{surv} itself is near zero, so the agent only collects a single episode's reward before collapse. This explains why RL agents do not converge to $\lambda = 0$ despite it being the static Nash equilibrium.

Separating Dynamic Structure from Activation Function. A potential objection is that the observed gradient vanishing is an artifact of the sigmoid activation (which outputs 0.5 when weights are small), rather than a game-theoretic property. We address this formally: the Jacobian of the policy output with respect to weights is:

$$\frac{\partial \lambda_i}{\partial W} = \sigma(z)(1-\sigma(z)) \cdot s_t^i \tag{9}$$

where $z = W^\top s_t^i + b$. At $W \approx 0, b \approx 0$, we have $\sigma(z) \approx 0.5$, giving $\sigma(1-\sigma) = 0.25$—the *maximum* of the sigmoid derivative, not a saturation region. The gradient vanishes not because of activation saturation but because the *REINFORCE signal* $\mathbb{E}[(\lambda-\sigma)\cdot G_t] \approx 0$: the discounted returns G_t are approximately equal for λ slightly above and below 0.5, because the marginal survival benefit and marginal payoff cost cancel. This is confirmed empirically across five independent validations: (i) replacing sigmoid with `tanh`, (ii) DNN ablation with 4 architectures (Appendix D), (iii) KPG with K=0,1,2 (Appendix E), (iv) scaling to $N = 100$, and (v) extending to spatial grid environments (Appendix F)—all producing identical convergence to $\lambda \approx 0.5$.

B Price Equation: Ecological Boundary Conditions

Theorem 1 requires the migration rate m to satisfy:

$$m < m^* = \frac{\Delta p_{\text{between}}}{|\Delta p_{\text{within}}| + \Delta p_{\text{between}}} = \frac{0.046}{0.115 + 0.046} = 0.286 \tag{10}$$

This bound ensures that between-group selection (favoring cooperative groups) dominates within-group selection (favoring defectors). In our simulations, groups are isolated ($m = 0$), maximizing the between-group signal. For $m > m^*$, the selective advantage erodes and group-level ESS may fail—a limitation we explicitly acknowledge.

C Sensitivity Analysis

Multiplier M. We vary $M \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$ with $N = 20$, Byz=30%, 10 seeds per condition. The Nash Trap ($\lambda \approx 0.5$ under selfish RL) and unconditional commitment optimality ($\phi_1^* = 1.0$) are preserved across $M \in [1.2, 2.0]$.

Recovery function $f(R_t)$ parameters. We sweep $R_{\text{crit}} \in \{0.10, 0.15, 0.20, 0.25\}$ and recovery rate scales $\in \{0.5\times, 1.0\times, 2.0\times\}$ (12 conditions, $N = 20$, Byz=30%, 5 seeds each). Across all 12 conditions:

- Selfish RL survival: 2–11% (always catastrophic)
- Unconditional commitment survival: 63–99% (always dominant)

Even under the most hostile configuration ($R_{\text{crit}} = 0.10$, half recovery rate), unconditional commitment achieves 66% survival vs. selfish RL's 11%. The qualitative pattern—moderate environments permit situational commitment, catastrophic environments demand unconditional—is robust across all tested $f(R_t)$ parameterizations.

D DNN Ablation: Network Depth vs. Nash Trap

To verify that the Nash Trap is game-theoretic rather than a function approximation limitation, we test four policy architectures with increasing depth (Table 7). All experiments use N=20, Byz=30%, 300 episodes $\times$ 5 seeds, with REINFORCE updates.

Table 7: DNN ablation: Nash Trap persists across all network depths (non-linear PGG, Byz=30%, last 30 episodes).

Architecture	$\bar{\lambda}$	**Survival %**	**Welfare**
Linear ($\sigma(w^\top s + b)$)	0.502	5.3	24.2
MLP-16 (1 hidden, ReLU)	0.500	5.3	24.2
MLP-32-16 (2 hidden, ReLU)	0.500	5.3	24.2
MLP-64-32-16 (3 hidden, ReLU)	0.500	5.3	24.2

All architectures converge to identical $\lambda \approx 0.500$ with 5.3% survival, confirming that the Nash Trap is invariant to policy network expressiveness. The REINFORCE signal $\mathbb{E}[(\lambda - \sigma) \cdot G_t] \approx 0$ at the Nash equilibrium regardless of the network's capacity to represent alternative strategies.

E KPG: K-Level Policy Gradient Comparison

We implement a simplified K-Level Policy Gradient (KPG) [12] to test whether recursive opponent anticipation escapes the Nash Trap (Table 8). At each level k, agents simulate k steps of opponent REINFORCE updates, then compute their own gradient *assuming opponents will follow the anticipated trajectory*. Experiments use N=20, Byz=30%, 300 episodes $\times$ 5 seeds.

Table 8: KPG fails to escape Nash Trap (non-linear PGG, Byz=30%, last 30 episodes).

K-Level	$\bar{\lambda}$	Survival %	Welfare	Time Ratio
K=0 (REINFORCE)	0.500	2.7	24.2	1.0×
K=1	0.500	7.3	24.2	1.2×
K=2	0.500	8.7	24.2	1.8×

The marginal survival improvement from K=0 to K=2 (2.7% $\rightarrow$ 8.7%) comes at 1.8× computational cost, while λ remains trapped at 0.500. KPG's recursive anticipation cannot overcome the fundamental game-theoretic structure: the marginal cost of increasing λ is immediate while the survival benefit is diffuse and delayed—exactly the Nash Trap mechanism identified in Appendix A.

F Spatial Social Dilemma Extension

To test whether the Moral Commitment Spectrum generalizes beyond scalar-state PGG, we construct a grid-based social dilemma with spatial structure (Table 9).

Environment. A 5×5 grid with N=15 agents (30% Byzantine). Each cell has an independent resource level $R_{x,y} \in [0, 1]$ with non-linear recovery dynamics identical to Eq. 4. Agents observe only their position, local resource, neighbor resource average, global average, and crisis flag ($R_{x,y} < R_{\text{crit}}$). At each step, agents choose both a commitment level $\lambda \in [0, 1]$ and a movement direction. Resource depletion is proportional to the number of agents at each cell and their free-riding intensity. Stochastic shocks (Poisson-distributed) and global degradation (0.2% per step) create sustained pressure. Collapse occurs when *global* average resource falls below 0.01.

Table 9: Spatial social dilemma (5×5 grid, N=15, Byz=30%, 200 ep $\times$ 5 seeds, last 30 episodes).

Method	$\bar{\lambda}$	Survival %	Welfare
Selfish RL (Nash Trap)	0.500	0.0	12.2
Free-riding (λ=0)	0.000	0.0	10.0
Situational	0.423	1.3	11.9
Unconditional (ϕ_1=1.0)	**1.000**	**34.7**	**14.4**

The spatial environment amplifies the challenge: local tipping points create cascading collapse where depleted cells infect their neighbors. Selfish RL exhibits the same Nash Trap ($\lambda = 0.500$) as in scalar PGG, confirming the phenomenon is environment-invariant. Situational commitment's crisis-reduction mechanism ($\lambda \rightarrow 0.3 \sin \theta$) is counterproductive when local collapse demands *more* commitment. Only unconditional commitment ($\phi_1 = 1.0$) achieves meaningful survival (34.7%), validating the Moral Commitment Spectrum in spatially-extended settings.